

MEDICONNECT PHASE-1

CLINICAL REPOSITORY IMPLEMENTATION GUIDE VERSION 1.0

OVERVIEW

This repository package contains the complete clinical knowledge layer for MediConnect.fit Phase-1.

The repositories are the source of truth for:

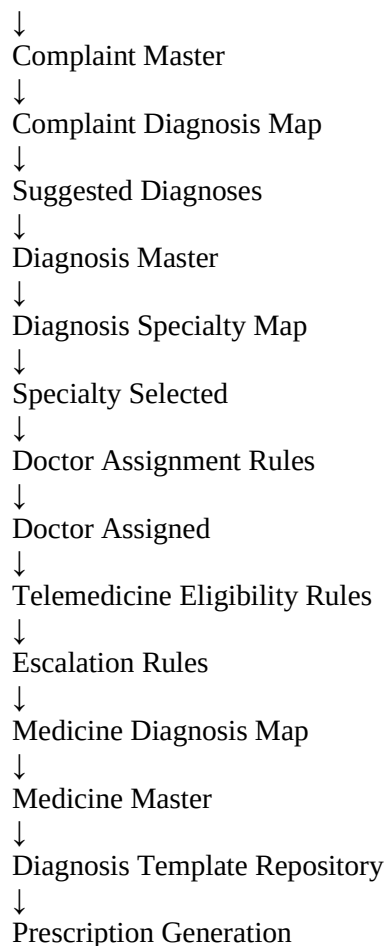
- Complaint intake
- Diagnosis suggestions
- Specialty routing
- Doctor assignment
- Telemedicine validation
- Escalation and safety checks
- Medicine recommendations
- Prescription generation

No clinical logic should be hardcoded inside application code.

All logic must be dynamically read from repository tables.

SYSTEM FLOW

PATIENT COMPLAINT



DATABASE IMPLEMENTATION

STEP 1

Create master tables for:

complaint_master
complaint_diagnosis_map
diagnosis_master
diagnosis_specialty_map

specialty_master
doctor_assignment_rules
medicine_master
medicine_diagnosis_map
telemedicine_eligibility_rules
escalation_rules
lab_test_master
diagnosis_template_repository

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STEP 2

Import repository data exactly as supplied.
Do not rename codes.
Do not modify IDs.
Do not remove records.

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STEP 3

Treat these as permanent business keys:

complaint_id
diagnosis_id
specialty_id
doctor_id
medicine_id
template_id

These IDs must never change.

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APPLICATION LOGIC

STEP 4

Complaint Mapping Engine

Input:

Patient complaint

Output:

Ranked diagnosis suggestions

Use:

Complaint Master

+

Complaint Diagnosis Map

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STEP 5

Diagnosis Routing Engine

Input:

Selected diagnosis

Output:

Mapped specialty

Use:

Diagnosis Specialty Map

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STEP 6

Doctor Assignment Engine

Input:

Specialty

Output:

Doctor

Use:

Doctor Assignment Rules

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STEP 7

Telemedicine Validation Engine

Input:

Diagnosis

Output:

Telemedicine Allowed / Not Allowed

Use:

Telemedicine Eligibility Rules

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STEP 8

Safety Engine

Input:

Complaint

Diagnosis

Output:

Emergency

Urgent

Routine

Use:

Escalation Rules

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STEP 9

Prescription Suggestion Engine

Input:

Diagnosis

Output:

Medicine Suggestions

Use:

Medicine Diagnosis Map

Medicine Master

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STEP 10

Prescription Template Engine

Input:

Diagnosis

Output:

Advice

Follow-up

Suggested Tests

Suggested Medicines

Use:

Diagnosis Template Repository

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API REQUIREMENTS

Create internal APIs for:

Complaint Search

Diagnosis Search

Specialty Search

Doctor Search

Medicine Search

Template Search

Safety Validation

Telemedicine Validation

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IMPORTANT RULES

DO NOT hardcode:

Diagnosis names

Specialties

Medicine mappings

Telemedicine rules

Escalation rules

Routing logic

Doctor assignment logic

All values must come from repository tables.

VERSION CONTROL

All repositories are frozen.

Future updates will be supplied through version-controlled repository releases.

Developer must support repository replacement without application code changes.

SUCCESS CRITERIA

When a patient enters a complaint:

Complaint

→ Diagnosis

→ Specialty

→ Doctor

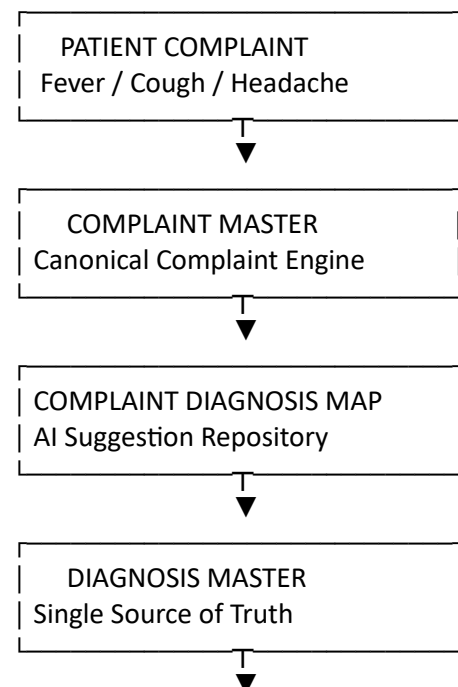
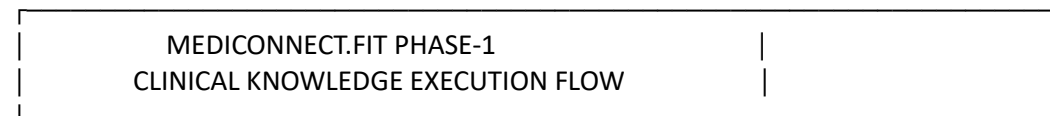
→ Telemedicine Validation

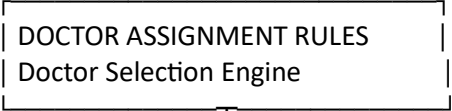
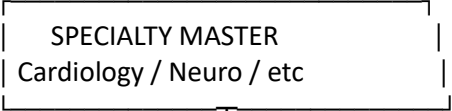
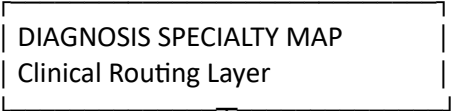
→ Escalation Validation

→ Medicine Suggestions

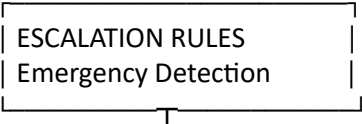
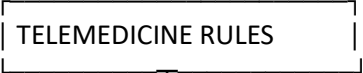
→ Prescription

must work entirely through repository-driven logic.

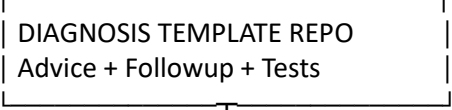
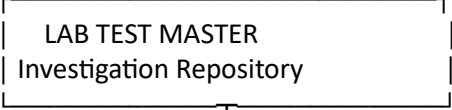
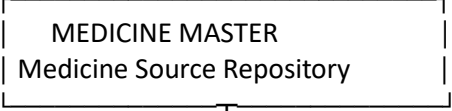
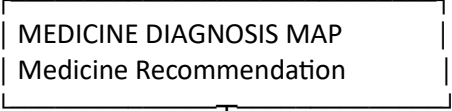




SAFETY VALIDATION LAYER



CLINICAL DECISION SUPPORT LAYER



PRESCRIPTION GENERATION
Final Doctor Output

ABDM / EHR / CONSULTATION
Patient Encounter Record



MEDCONNECT.FIT PHASE-1 CLINICAL KNOWLEDGE ENGINE

REPOSITORY DRIVEN CLINICAL ARCHITECTURE & DATA FLOW

END-TO-END FLOW: FROM PATIENT COMPLAINT TO PRESCRIPTION & HEALTH RECORD

VERSION 1.0 RELEASE MAY 2025

ARCHITECTURE PRINCIPLES

- NO CLINICAL LOGIC MAY BE HARDCODED.
- ALL MAPPINGS MUST COME FROM REPOSITORY TABLES.
- ALL SPECIALTY ROUTING MUST COME FROM REPOSITORY TABLES.
- ALL TELEMEDICINE DECISIONS MUST COME FROM REPOSITORY TABLES.
- ALL ESCALATION DECISIONS MUST COME FROM REPOSITORY TABLES.
- ALL MEDICINE RECOMMENDATIONS MUST COME FROM REPOSITORY TABLES.
- REPOSITORY IDS ARE IMMUTABLE BUSINESS KEYS.
- REPOSITORY REPLACEMENT MUST NOT REQUIRE CODE CHANGES.
- ALL CLINICAL DECISIONS ARE DATA DRIVEN & AUDITABLE.
- SUPPORTS SCALABILITY, MAINTAINABILITY & VERSIONING.
- COMPLIANT WITH ABDM, NDHM, HIPAA & DATA PRIVACY.
- BUILT FOR FUTURE AI / ML ENHANCEMENTS.

LEGEND (SWIMLANES)

- 1 Patient Intake Layer
- 2 Clinical Core Layer
- 3 Safety & Validation Layer
- 4 Clinical Decision Support Layer
- 5 Execution Layer
- 6 Governance & Infrastructure Layer



GOVERNANCE, CONTROL & INFRASTRUCTURE LAYER



REPOSITORY VERSION CONTROL

- Version Management (v1.0, v1.1, v1.2...)
- Change Tracking & Release Management
- Rollback & Audit History
- Effective From / Effective To



REPOSITORY ADMIN CONSOLE

- Upload Repository
- Validate Data
- Approve / Activate / Deactivate
- Audit Logs & Change History
- Role Based Access Control



AUDIT & TRACEABILITY ENGINE

- Track Every Decision
- Complaint → Diagnosis → Specialty → Doctor → Medicine → Prescription
- Who, When, What, Why (Full Trace)
- Immutable Audit Logs



KEY INTEGRATIONS

- Doctor Management System
- Pharmacy System
- Laboratory System
- Payment & Billing System
- ABDM / NDHM / PHR
- Notification & Communication

BENEFITS

- Standardized Clinical Decision Making
- Consistent & Safe Patient Care
- Scalable Repository Driven Architecture
- Easy Updates Without Code Changes
- Fully Audit-able & Compliant

END TO END
CLINICAL FLOW



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